

HOMEWORK 3

LEARNING THEORY, KERNELS AND SVMs

CMU 10-701: MACHINE LEARNING (FALL 2024)

<https://piazza.com/cmu/fall2024/10701/home>

OUT: Wednesday, October 9th, 2024

DUE: Monday, October 28th, 2024, 11:59pm

START HERE: Instructions

- **Collaboration policy:** Collaboration on solving the homework is allowed, after you have thought about the problems on your own. It is also OK to get clarification (but not solutions) from books or online resources, again after you have thought about the problems on your own. There are two requirements: first, cite your collaborators fully and completely (e.g., “Jane explained to me what is asked in Question 2.1”). Second, write your solution *independently*: close the book and all of your notes, and send collaborators out of the room, so that the solution comes from you only. See the Academic Integrity Section in our [Course Syllabus](#) for more information:
- **Late Submission Policy:** See the late homework policy in our [Course Syllabus](#).
- **Submitting your work:**

- **Gradescope:** Submit your homework to the Homework 3: Written Gradescope submission slot.

Please use the provided template. The best way to format your homework is by using the Latex template released in the handout and writing your solutions in Latex. However submissions can be handwritten onto the template, but should be labeled and clearly legible. If your writing is not legible, you will not be awarded marks. Each derivation/proof should be completed in the boxes provided below the question, **you should not move or change the sizes of these boxes** as Gradescope is expecting your solved homework PDF to match the template on Gradescope. If you find you need more space than the box provides you should consider cutting your solution down to its relevant parts, if you see no way to do this, please add an additional page at the end of the homework and guide us there with a ‘See page xx for the rest of the solution’.

Regrade requests can be made after the homework grades are released, however this gives the TA the opportunity to regrade your entire paper, meaning if additional mistakes are found then points will be deducted.

For multiple choice or select all that apply questions, shade in the box or circle in the template document corresponding to the correct answer(s) for each of the questions. For $\text{\LaTeX}$ users, use $\blacksquare$ and $\bullet$ for shaded boxes and circles, and don’t change anything else. If an

answer box is included for showing work, **you must show your work!**

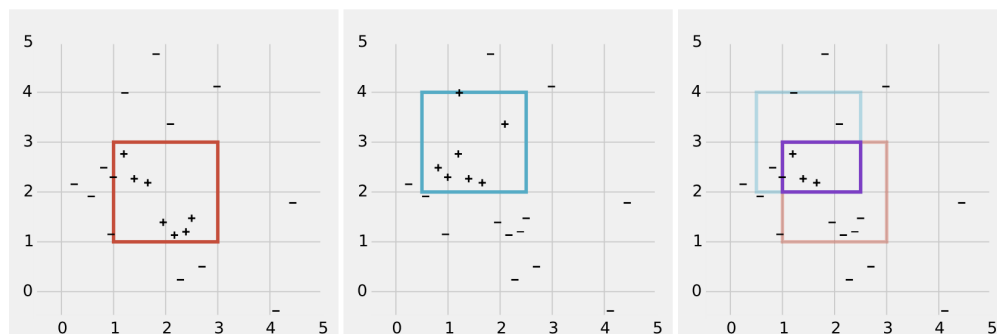
1 VC Dimension [52 Points]

Recall that we call a set of points shattered by a class of functions H if all possible $\{-1, +1\}$ labelings of the points can be produced by some function in H . The Vapnik-Chervonenkis (VC) dimension is the size of the largest set of points that can be shattered by the hypothesis space. See the lecture notes, and recording for more information.

1.1

In this problem, we will explore the hypothesis space where each hypothesis is a combination of two simpler hypotheses. More precisely, given two hypotheses h_1 and h_2 , we define $h = h_1 \cap h_2$ as a new hypothesis that labels an example $+1$ only if both h_1 and h_2 give the label $+1$, otherwise, it is labeled -1 . We can extend this to *sets* of hypotheses: given two sets of hypotheses H_1 and H_2 , define $H^* = \{h_1 \cap h_2 : h_1 \in H_1, h_2 \in H_2\}$ as the set of all intersections of hypothesis pairs from the two classes H_1 and H_2 .

As an example, let H_1 be the set of classifiers in $\mathbb{R}$ that assigns the label $+1$ if the example is larger than some threshold a . Let H_2 be the set of classifiers in $\mathbb{R}$ that assigns the label $+1$ if the example is smaller than some threshold b . Then H^* would be the set of all intervals (a, b) in $\mathbb{R}$ that assigns $+1$ if the example is inside the interval. Another example is when H_1 and H_2 is the set of all (axis-aligned) squares in $\mathbb{R}^2$, H^* is the set of all axis-aligned rectangles. This example is illustrated below. On the left, we have a single square classifier h_1 ; in the middle we again have a square classifier h_2 ; and on the right, we have $h_1 \cap h_2$, which is a rectangle classifier.



Keep in mind that these are only examples. We are looking for results that can apply generally to any pair of hypotheses classes.

1. **[15 pts]** Suppose that the *shattering coefficient* of H_1 is $H_1[n]$ (i.e. the maximum number of ways that the hypothesis space H_1 can label a set of n points is $H_1[n]$). Similarly, suppose that the shattering coefficient of H_2 is $H_2[n]$. Show that $H^*[n] \leq H_1[n]H_2[n]$.

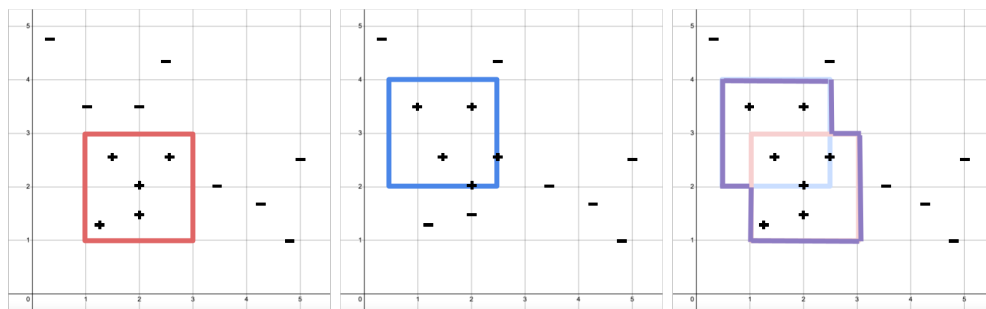
2. **[5 pts]** Show that if the VC dimension of a hypothesis space H is d , then the shattering coefficient $H[d]$ is 2^d .

3. [15 pts] Let H be a hypothesis space with VC dimension d . Define H^* as the hypothesis space produced by all intersections of pairs of hypotheses from H (assuming that $H_1 = H_2 = H$ in our above definitions). Use your results from Question 1.2 to show that the VC dimension d_* of H^* is bounded by $O(d \log d)$. **Hint:** You may find the following fact useful; if $2^x \leq x^y$, then $x \leq k \cdot y \log y$ for some constant k . **Hint:** Since d_* is the VC dimension of H^* , then by definition, there exists a set S of d_* points that is shattered by H^* . By Sauer's lemma, we know that the maximum number of ways that H can label S is bounded by $O(d_*^d)$. That is, $H[d_*] = O(d_*^d)$.

1.2

Now, we will explore the hypothesis space where each hypothesis is a union of two simpler hypotheses. More precisely, given two hypotheses h_1 and h_2 , we define $h = h_1 \cup h_2$ as a new hypothesis that labels an example $+1$ if either h_1 or h_2 give the label $+1$, otherwise, it is labeled -1 . We can extend this to *sets* of hypotheses: given two sets of hypotheses H_1 and H_2 , define $\tilde{H} = \{h_1 \cup h_2 : h_1 \in H_1, h_2 \in H_2\}$ as the set of all unions of hypothesis pairs from the two classes H_1 and H_2 .

As an example, let H_1 be the set of classifiers in $\mathbb{R}$ that assigns the label $+1$ if the example is less than or equal to some threshold a . Let H_2 be the set of classifiers in $\mathbb{R}$ that assigns the label $+1$ if the example is greater than or equal to some threshold b . Then $\tilde{H}$ would be the set of all values not in the interval (a, b) in $\mathbb{R}$ that assigns $+1$ if the example is outside the interval. Another example is when H_1 and H_2 is the set of all (axis-aligned) squares in $\mathbb{R}^2$, $\tilde{H}$ is the set of all the unions of the squares in H_1 and H_2 . This example is illustrated below. On the left, we have a single square classifier h_1 ; in the middle we again have a square classifier h_2 ; and on the right, we have $h_1 \cup h_2$, the union of the two squares.



Keep in mind that these are only examples. We are looking for results that can apply generally to any pair of hypotheses classes.

1. [10 pts] Suppose that the *shattering coefficient* of H_1 is $H_1[n]$ (i.e. the maximum number of ways that the hypothesis space H_1 can label a set of n points is $H_1[n]$). Similarly, suppose that the shattering coefficient of H_2 is $H_2[n]$. Show that $\tilde{H}[n] \leq H_1[n]H_2[n]$.

2. [7 pts] Let H be a hypothesis space with VC dimension d . Define $\tilde{H}$ as the hypothesis space produced by all union of pairs of hypotheses from H (assuming that $H_1 = H_2 = H$ in our above definitions). Show that the VC dimension $\tilde{d}$ of $\tilde{H}$ is bounded by $O(d \log d)$. **Hint:** You may find the following fact useful; if $2^x \leq x^y$, then $x \leq k \cdot y \log y$ for some constant k .

2 SVM Decision Boundaries [12 pts]

Figure 1 shows SVM decision boundaries resulting from different combinations of kernels, slack penalties, and RBF bandwidths. Match each plot from Figure 1 with one of the SVM settings.

To obtain full credit, you do not need to provide any justification, but only provide the correct pairing.

RBF kernel: $K(\mathbf{x}, \mathbf{z}) = \exp(-\gamma \|\mathbf{x} - \mathbf{z}\|_2^2)$

1. Linear kernel, $C = 0.1$
2. Linear kernel, $C = 100$
3. RBF kernel, $C = 1$, $\gamma = 10$
4. RBF kernel, $C = 1$, $\gamma = 50$
5. RBF kernel, $C = 0.1$, $\gamma = 0.1$
6. RBF kernel, $C = 20$, $\gamma = 0.1$

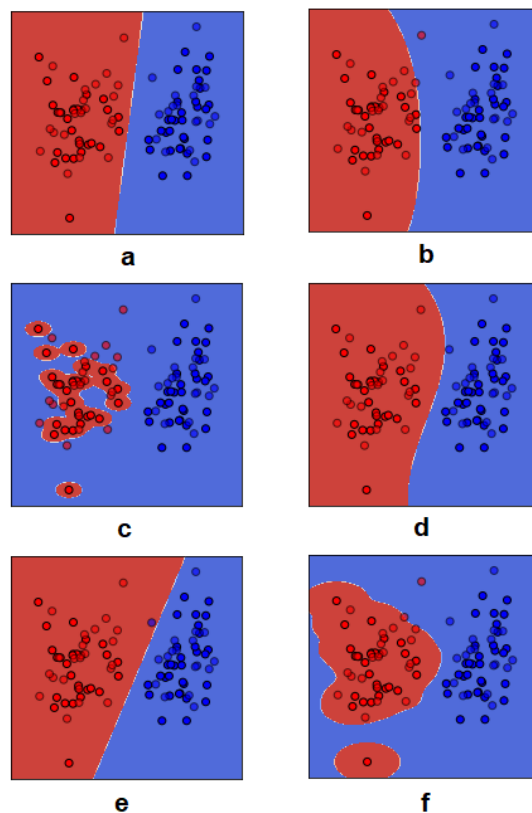


Figure 1: SVM decision boundaries learned with different parameter settings.

3 Collaboration Questions

1. (a) Did you receive any help whatsoever from anyone in solving this assignment?

SOL: Yes / No.

- (b) If you answered ‘yes’, give full details (e.g. “Jane Doe explained to me what is asked in Question 3.4”)

SOL:

2. (a) Did you give any help whatsoever to anyone in solving this assignment? **SOL:**

Yes / No.

- (b) If you answered ‘yes’, give full details (e.g. “I pointed Joe Smith to section 2.3 since he didn’t know how to proceed with Question 2”)

SOL:

3. (a) Did you find or come across code that implements any part of this assignment?

SOL: Yes / No.

- (b) If you answered ‘yes’, give full details (book & page, URL & location within the page, etc.).

SOL: